

32 An Overview of Animal Diversity

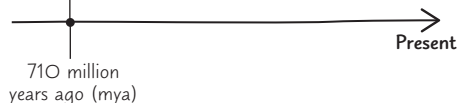
KEY CONCEPTS

- 32.1** Animals are multicellular, heterotrophic eukaryotes with tissues that develop from embryonic layers p. 674
- 32.2** The history of animals spans more than half a billion years p. 675
- 32.3** Animals can be characterized by body plans p. 679
- 32.4** Views of animal phylogeny continue to be shaped by new molecular and morphological data p. 682

Study Tip

Draw a time line: Create a time line like the one shown here, and mark when the following occurred: first biochemical evidence of animals, Edicaran biota, Cambrian explosion, colonization of land by vertebrates, and dinosaur extinction. Briefly describe the significance of each event.

First biochemical evidence of animals:
Sediments containing steroids from sponges



➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 32
- Interview with Nicole King: Investigating the ancestry of choanoflagellates

For Instructors to Assign (in Item Library)

- Tutorial: Visualizing Animal Body Symmetry and Axes
- Tutorial: Animal Body Plans
- Scientific Skills Exercise: Calculating and Interpreting Correlation Coefficients



Figure 32.1 This chameleon can wield its long, sticky tongue with blinding speed to capture its prey. Other predatory animals use strength, concealed traps, or toxins to capture prey. Likewise, herbivorous animals can strip plants bare of leaves or seeds, while parasitic animals weaken their hosts by consuming their tissues or body fluids. Overall, we can think of animals as a kingdom of consumers.

What key characteristics of animals make them such efficient consumers?

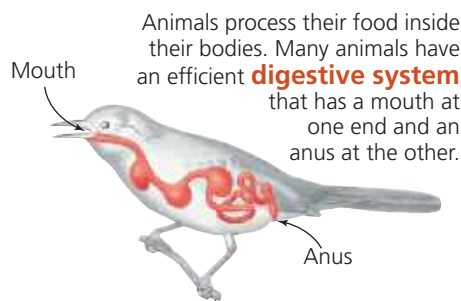


Animals are **heterotrophs**: They obtain energy and nutrients by eating other organisms.



Animal embryo (cross section)

Animals have **tissues**, collections of specialized cells that function as a unit. Tissues form from layers of embryonic cells.

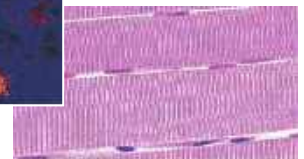


Animals process their food inside their bodies. Many animals have an efficient **digestive system** that has a mouth at one end and an anus at the other.

Nerve cell



The tissues of animals include those composed of unique **nerve and muscle cells**.



Muscle cells



Their nerve and muscle cells enable animals to **move** as well as to **detect and capture** potential prey.